

AI for IoT-NDN: Enhancing IoT with Named Data Networking and Artificial Intelligence

Mohamed Ahmed M. Hail
Institute of Telematics
University of Lübeck
Ratzeburger Allee 160
Lübeck, Germany
m.hail@uni-luebeck.de

Ali Abdulqader Bin-Salem
School of Computer Sci. and Tech.
Zhoukou Normal University
Zhoukou 466001
Henan, China
ali@zknv.edu.cn

Waddah Munassar
Faculty of Computer Science and IT
University of Aden
Aden, Yemen
w_munassar@yahoo.com

Abstract—This paper presents a comprehensive study on the integration of Artificial Intelligence (AI) methodologies within the rapidly evolving Internet of Things (IoT) landscape, particularly focusing on Named Data Networking (NDN). As IoT permeates various sectors like Smart Homes, Cities, Grids, and Health, NDN emerges as a robust solution for managing the extensive data flow, addressing key aspects such as naming, mobility, data aggregation, and security. Despite NDN's efficacy, its implementation confronts challenges related to high operational costs, influenced by bandwidth usage and network popularity. Our research investigates how AI can enhance NDN's functionality in IoT contexts, aiming to optimize data dissemination and network efficiency. The study explores AI's potential in bolstering IoT capabilities within the NDN framework, offering insights into future directions for IoT-based NDN networks.

Index Terms—Internet of Things (IoT), Named Data Networking (NDN), Artificial Intelligence (AI)

I. INTRODUCTION

The Internet of Things (IoT) connects everyday objects to the Internet, allowing them to communicate and share data. Characterized by compactness, low power consumption, and integrated components like sensors and actuators, IoT faces challenges such as limited sensor power, bandwidth constraints due to high data volumes, and significant security concerns. Addressing these issues through effective management of power, communication, data dissemination, and security is crucial [1].

Named Data Networking (NDN), a content-centric alternative to the host-centric Internet architecture, is well-suited for IoT applications, offering benefits like reduced power consumption and enhanced security [2], [3]. NDN's advantages in

IoT include improved scalability and security, but require new optimization techniques [4]. The integration of AI with IoT-based NDN, as suggested by [5], could enhance these networks by predicting traffic patterns and optimizing data transmission. This research investigates AI techniques in IoT-based NDN, aiming to support various applications.

Integrating Artificial Intelligence (AI) with the Internet of Things (IoT) in the context of Named Data Networking (NDN) represents a critical evolution in addressing the challenges of our increasingly connected world, particularly in managing the vast data generated by IoT. This integration is not just a technological leap but a necessary response to the unprecedented surge in data volume, diversity, and velocity, which poses significant challenges in data management, security, and network resource utilization. AI's role in this framework is pivotal, as it significantly enhances data processing, enabling smarter, more adaptive networks capable of dynamic decision-making and real-time analysis, crucial in sectors such as Smart Cities, Healthcare, and Industrial Automation. Furthermore, AI helps address IoT's security and privacy concerns by introducing advanced algorithms for anomaly detection, encryption, and secure data sharing. The integration of AI with IoT-NDN drastically improves network scalability and sustainability, a necessity as the proliferation of IoT devices intensifies network strain. This is especially crucial in scenarios with limited bandwidth, where AI can optimize resource allocation, reduce congestion, and enhance system resilience. In essence, AI's integration into IoT-NDN is indispensable for overcoming modern networking complexities and paving the way for

smarter, more efficient network solutions.

The paper is structured into several sections. I sets the context, highlighting the significance of AI in IoT-NDN. II surveys existing research, while III explores their fundamental principles. Section IV details various AI methodologies. In Section V, we explore examples and applications of these use cases. Additionally, Section VI discusses the future challenges associated with integrating AI into IoT-NDN. The VII concludes the paper, summarizing key findings and discussing AI's role in advancing IoT-NDN.

II. RELATED WORK

This section focuses on the current advancements in the integration of AI with the IoT-NDN. The rationale for selecting this topic is in the extensive body of research available in this domain. The combination of IoT-NDN-AI has several possibilities, including the study of providing a complete description of the developing technologies and applications of NDN [6], the development of mobility in IoT-NDN [7], the inherent caching capabilities of ICN in IoT [8], and so on. This paper provides a comprehensive analysis of several approaches and their respective domains. The objective of the study conducted in [6] is to provide a complete description of the developing technologies and applications of NDN. This is achieved by analyzing and examining research articles released between the years 2018 and 2022. The NDN-AI section reveals that within the context of IoT edge computing, the absence of content identifiers results in a greater emphasis on host-centric communication, hence highlighting the need for NDN-AI. The implementation of a machine algorithm designed for user-centric communication has been proposed as a potential solution. Additionally, the evolution of NDN-AI incorporates edge computing as a means to address the challenges posed by large input data and minimize latency delays. This approach also enhances security and stability in routing by mitigating the risk of a collusive Interest Flooding Attack (CIFA), which involves the malicious generation of Interest packets by producers within the NDN traffic. Researchers in [7] are directing their attention toward the convergence of cloud IoT (C-IoT) and NDN-AI-IoT. This research offers a detailed examination of the technologies, specifically focusing on their methodological and parameter explanations. Three approaches are used to do this: KITE, a producer mobility support scheme (PMSS), and hybrid network mobility (hybrid NeMO). The mathematical

formulae for each approach are derived from the mobility scenario in order to quantify the handover delay and signaling cost. The hybrid NeMO system offers a binding information table as a replacement for the base function of information forwarding. It employs AI to enhance the data processing capabilities of the mobile producer, hence expediting the transmission process. Additionally, all relevant information is consistently updated via the use of the forwarding information base (FIB) in NDN. The findings demonstrate the effectiveness of hybrid NeMO in reducing signaling cost and delay compared to other approaches. Safitri et al. [9] examined the use of artificial AI techniques in the classification of names within the context of information-centric networking (ICN) for the IoT. A hybrid ICN was developed via the use of machine learning methodologies in order to address the requirements of a pragmatic IoT technology. The algorithms of supervised learning, unsupervised learning, and reinforcement learning underwent evaluation prior to their selection as the content-forwarding technique. The empirical findings indicate that the extended learning classifier system exhibits superior performance compared to other algorithms while using the reinforcement learning scheme. However, it is possible to use various machine learning approaches such as logistic regression, nearest neighbors, decision trees, and naive Bayes as reference points for this experiment. The authors Pahl et al. [8] presented a proposal to integrate AI into ICN for the IoT. This integration is achieved by using the inherent caching capabilities of ICN in conjunction with the data-centric nature of IoT. Additionally, the authors suggested the use of machine learning algorithms to determine the most suitable caching method for individual data items and to enhance the accuracy of node behavior prediction. Nevertheless, the determination of criteria such as performance and scalability remains pending until the implementation phase. In ICN-IoT, Divya and colleagues [10] suggested an upgraded content caching solution for heterogeneous IoT architecture. AI-based collaborative filtering is used in the edge cloud. To control cloud database traffic, the suggested content caching technique uses collaborative filtering to strategically cache data on edge nodes. The evaluations compared the suggested technique to different benchmark methodologies. Analysis shows that the proposed method surpasses the best solutions in cache hit ratio, content retrieval time, and average hop count. However, the recommended

technique must be tested on various NDN architectures. Caching strategies must take into account a wide range of parameters associated with the requested content and the different caching capabilities of all edge nodes to enable a network that can accommodate growth and reduce material retrieval time. Internet of Medical Things (IoMT) is a key tool in reducing and managing COVID-19. Khan et al. [11] offers a Smart COVID-19 Pandemic-Controlled Eradication Mechanism utilizing NDN and IoT (SPICE-IT), a revolutionary virus-fighting method using NDN-IoT technology. SPICE-IT uses pull-based reporting to autonomously monitor indoor surroundings and detect transgressions on local and cloud servers. Each individual undergoes assessment using an advanced system that combines intelligent face mask tracking and temperature measuring. The cloud server governs the response mechanism of the central system via the use of an adaptive AI system. The use of a caching method that relies on lengthy short-term memory effectively mitigates the occurrence of cache overflow and reduces the overall congestion within the network.

III. IoT-NDN AND AI INTEGRATION

A. IoT-NDN Overview

An overview of the IoT-NDN architecture is provided in this section; a further description can be found in [12]. Previous research articles, including [13] and [14], have covered the implementation of IoT-NDN. The main elements of the IoT-NDN and their functionalities will be briefly introduced in this section. All the parts of the IoT-NDN system are visually summarized in Figure 1.

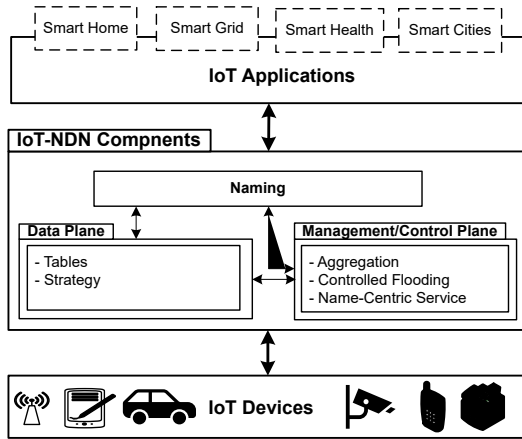


Fig. 1: IoT-NDN System Architecture and its Components

The IoT-NDN device layer, which includes all heterogeneous IoT-NDN devices, is located at the bottom of the IoT-NDN architecture. The IoT-NDN applications layer, which requests the data, sits on top of the IoT-NDN architecture. Whereas a networking layer called IoT-NDN is responsible for sending the desired data from the underlying layer to the overlaying layer.

The IoT-NDN architecture has three fundamental components, namely the Naming, Data Plane, and Management and Control Plane. The Naming component encompasses a distinctive naming framework and methodology specifically designed for wireless network devices operating under limitations and restrictions. The Management and Control Plane encompasses many subcomponents, including the Alias Service, Controlled Flooding, Unicast Faces, Forwarding Services, Intra Node Protocol (INP), and Controlled Flooding. The Data Plane component encompasses the caching and forwarding techniques used for IoT-NDN data. The IoT-NDN components control the naming, caching, and techniques for IoT-NDN packets in the system. The requirements of IoT-NDN were taken into consideration when designing, developing, and adding each component shown in Figure 1.

The CS, PIT, and FIB tables are present in every device that makes use of the IoT-NDN architecture. Faces are used to receive and send all IoT-NDN packets, including Interest and Data packets. IoT-NDN use faces to streamline the processing and routing of messages pertaining to the network or local applications in the IoT-NDN system. [12] presents additional details regarding the IoT-NDN components and their roles. A detailed description of how faces are implemented in IoT-NDN may be found in [12].

B. Integration of AI for IoT-NDN

Artificial intelligence (AI) has the potential to greatly benefit the Internet of Things (IoT) and Named Data Networking (NDN). IoT-NDN connects various devices that share data over the internet, and NDN is a network architecture that focuses on content retrieval.

IoT-NDN devices have become an integral part of our lives, offering convenience and generating a vast amount of data related to user behavior and preferences. However, many businesses struggle with how to handle and process this enormous volume of data, which can hinder the growth and potential of IoT and NDN, and also optimize the energy consumption.

AI can play a crucial role in managing and analyzing the data generated by IoT and NDN devices. It enables us to extract valuable insights, optimize data retrieval, and improve overall network performance while simultaneously helping to optimize energy consumption.

The AI algorithms and techniques component has been added to the IoT-NDN architecture (see Figure 2). In the other sections, we delve deeper into the various AI techniques and their applications within IoT-NDN, illustrating how they enhance caching strategies, forwarding and routing decisions, and energy consumption optimization to harness the full potential of this dynamic convergence.

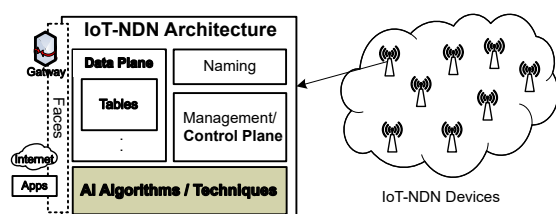


Fig. 2: IoT-NDN system architecture, including the AI Complement

IV. AI TECHNIQUES FOR IoT-NDN

AI approaches are crucial in the field of IoT-NDN for optimizing important system components such as caching strategies, forwarding mechanisms, routing selections, and energy usage. In order to improve the overall effectiveness and performance of the IoT-NDN, this section examines how AI contributes to each of these areas.

Caching Techniques: Content Prediction Based on Machine Learning: AI-driven models, such as machine learning algorithms, can forecast which data objects are most likely to be requested by IoT-NDN devices. These models make it possible to cache intelligent material by looking at historical data access patterns and context. Due to the network's easy access to commonly requested data, this improves user experience by reducing latency in data retrieval. Dynamic Cache Management: Based on real-time data flow patterns, reinforcement learning systems can continuously change caching tactics. While avoiding storage waste, this dynamic technique makes sure that the most pertinent and frequently visited content stays in the cache.

Mechanisms for Forwarding: Deep learning models, in particular deep neural networks, can

help with decisions regarding content-based routing. Deep learning algorithms can choose the best route for data forwarding by examining the properties of the contents. For instance, deep learning can determine the most effective path for real-time traffic data to take in order to efficiently reach its destination in a smart city scenario. Reinforcement Learning for Dynamic Forwarding: By taking into account network conditions and device needs, reinforcement learning agents can optimize forwarding methods. These agents acquire the ability to instantly modify routing choices in response to shifting traffic patterns and network congestion.

Route Selection: Routing optimization driven by machine learning: To make the best routing choices, machine learning models can examine network performance metrics, device locations, and historical data transfer trends. This guarantees that data is sent through the fastest route possible, cutting down on delays and network congestion. Predictive Routing: Based on the behavior of IoT-NDN devices, AI algorithms may forecast future data transfer requirements. Data is pre-positioned along predicted pathways thanks to predictive routing, which helps to minimize retrieval delays and lighten the load on the network.

Optimization of Energy Consumption: Energy-efficient IoT-NDN Reinforcement Learning: Reinforcement learning agents can optimize IoT-NDN device operations in energy-constrained contexts. These agents adapt their behavior to increase device longevity while retaining vital functionality. They learn when to send data, switch to low-power modes, or modify their activity.

AI approaches can be included into IoT-NDN architecture to provide dynamic, data-driven, and power-efficient network operations. In order to fully realize the potential of this convergence, these strategies allow IoT-NDN systems to adapt to changing circumstances, improve user experiences, and maximize resource consumption.

The table I provides a concise summary of AI algorithms and techniques related to the research areas of caching strategies, forwarding mechanisms, routing decisions, and energy consumption optimization in the context of IoT-NDN. It's important to note that there are also other techniques that may be of interest for research in the field of IoT-NDN. However, due to space limitations in this paper, we focus on these key techniques.

TABLE I: Overview of AI Techniques for IoT-NDN

AI Techniques	Caching Strategies	Forwarding Mechanisms	Routing Decisions	Energy Consumption Optimization
Machine Learning	- Predictive caching reduces latency in a smart city's traffic monitoring. - Adaptive cache management.	- Content-based routing using deep learning models. - Adaptive routing decisions.	Optimizes routing for IoT-NDN applications based on device behavior and network conditions.	Reinforcement learning for energy-efficient IoT-NDN device operations.
Reinforcement Learning	- Dynamic cache management. - Optimizes routing for delivery IoT-NDN Data, reducing delivery times.	Real-time adaptive forwarding.	- Real-time routing optimization. - Optimizes IoT-NDN device operations in a smart grid, extending device lifespan.	IoT-NDN device energy optimization.

V. USE CASES AND APPLICATIONS

In the context of deploying AI within the IoT-NDN framework, IoT devices can leverage the benefits of NDN to efficiently retrieve data from the nearest provider through a single data communication process. This is particularly advantageous in the context of existing edge techniques. In addition, numerous IoT applications, such as video surveillance, autonomous driving, smart home appliances, and industrial automation, derive significant advantages from the incorporation of AI capabilities, including data, picture, audio, and video analysis. It is clear that the integration of AI enables enterprises to swiftly respond to situations or take action due to the accelerated processing of data, which provides enhanced insights.

1) Smart Grids:

AI optimizes Smart Grid operations, enhancing outage alerts and power efficiency. Information-Centric Networking (ICN) improves data distribution in Smart Grids, with projects like C-DAX utilizing ICN for secure, scalable networks in renewable energy integration. ICN also supports retail energy transactions and Synchrophasor deployment in Smart Grids.

2) Smart Healths:

IoT applications in healthcare reduce costs and improve service accessibility. However, IP-based data transfer in healthcare poses security and privacy issues. NDN offers a solution with efficient content distribution and enhanced security. The PHINet framework

integrates Health-IoT with cloud computing for content-centric applications, and AI technology supports real-time data analysis while maintaining privacy in the Internet of Medical Things.

- 3) Smart Homes: IoT-enabled smart homes include devices like thermostats, sensors, and security systems. ICN is being researched as a communication solution for smart homes due to its efficient naming and security features. ICN provides improved communication security and reliability in smart homes. Examples include NDOMUS for intelligent residential environments and NDN-LC, a framework for lighting management in Building Automation Systems.

VI. FUTURE CHALLENGES

Incorporating Artificial Intelligence (AI) into the Internet of Things (IoT) through Named Data Networking (NDN) significantly boosts the efficiency of data retrieval in IoT devices, particularly enhancing systems like smart home applications, smart cities etc. with advanced analytical capabilities. To fully harness AI's potential in the IoT-NDN, it is essential to tackle its inherent challenges. This section is dedicated to exploring these key issues, presenting viable solutions, and emphasizing the continuous importance of research and development.

Limited Resource Environments: Challenge: The internet faces increasing traffic and data rates, especially in video and streaming, stressing the cur-

rent infrastructure. Solution: Develop lightweight, resource-efficient AI models, using edge AI for local processing. Explore federated learning for collaborative AI model training on IoT-NDN devices.

Data Security and Privacy: Challenge: IoT-NDN environments are prone to security risks due to sensitive data collection. Solution: Implement strong data encryption and privacy-preserving AI algorithms. Research: Focus on federated learning and privacy-preserving machine learning, and test these approaches in realistic topologies.

Network overhead and Scalability: Challenge: AI integration in IoT-NDN can increase network traffic and scalability issues. Solution: Optimize data transmission with intelligent data aggregation and enhance AI for distributed processing. Improve content caching in IoT-NDN using AI-driven collaborative filtering. Research: Investigate distributed AI models for collective decision-making without overloading the network.

Energy Efficiency: Challenge: IoT-NDN networks face high computational burden and power consumption due to content caching. Solution: Utilize dynamic power management, low-power AI hardware, and improve AI algorithms' energy efficiency. Explore AI for device control and data analysis with a focus on energy conservation.

Real-Time Prerequisites: Challenge: Some IoT-NDN applications require real-time AI responses, but face latency issues. Solution: Implement edge computing for real-time AI processing and use AI hardware accelerators for low-latency operations. Develop AI methods for rapid processing to meet real-time demands.

VII. CONCLUSION

In this paper, we've explored how AI enhances IoT-NDN, underlining its pivotal role in optimizing caching, routing, Forwarding, and energy efficiency. AI-driven IoT-NDN systems offer improved efficiency, scalability, and transformative impacts across industries, including healthcare, smart homes, and smart cities.

However, challenges like resource constraints, data privacy, and real-time requirements demand ongoing research and innovative solutions. As IoT-NDN continues to evolve with AI, it brings industry transformations, advancements in IoT applications, and environmental benefits.

The goal is to conduct experiments in simulated and real-world testbeds to validate and quantify the real-world impact of the suggested approaches in IoT-NDN-based AI. This practical implementation

and experimentation will provide valuable insights and further refine AI-driven IoT-NDN systems for real-world deployment.

In summary, AI is integral to IoT-NDN's evolution, promising a more connected, efficient, and sustainable future. Achieving the full potential of this transformative integration requires responsible development, robust privacy protection, and empirical validation through experiments.

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